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Learning Cardiac Electrophysiology with Graph Neural Networks for Fast Data-driven Personalised Predictions

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Abstract

Efficient modeling of cardiac electrophysiology is essential for advancing personalized medicine and improving treatment strategies. Traditional mathematical models provide accurate simulations but are computationally expensive, limiting their scalability to complex geometries and their generalization to patient-specific applications. In this work, we propose AGATA, an autoregressive graph neural network-based approach for simulating cardiac action potential propagation.

AGATA is trained on simple ellipsoid meshes yet demonstrates a remarkable ability to generalize to more complex and realistic geometries. It has been applied to meshes obtained from medical images, including one of the left ventricle endocardium (surface) and one of the two ventricles (volume). AGATA effectively captures spatial propagation in healthy tissue and reproduces patterns in damaged areas. The mean absolute error (MAE) between the values generated by the Finite Element Method and AGATA is 0.00003 (dimensionless) for a simple ellipsoid, and 0.007 for more complex geometries.

Unlike traditional methods, AGATA’s ability to adapt to previously unseen, intricate geometries without requiring retraining emphasizes its potential for patient-specific modeling. Additionally, AGATA achieves up to a twelve-fold reduction in computational time compared to the Finite Element Method, making it suitable for near-real-time applications while maintaining a good approximation of the electrical signal.

1 Introduction

Cardiovascular disease remains a leading global cause of mortality, with arrhythmias being a significant contributor. Identifying the regions responsible for arrhythmias is crucial, especially as they vary significantly between patients. While methods exist to detect scars [1], pointing out the precise areas triggering arrhythmia remains challenging, particularly as it needs to be adaptive to different patients. Traditional clinical approaches, such as electroanatomical mapping, are time-consuming and lack predictive capabilities.

Digital twin models simulate cardiac Action Potential (AP) propagation for protocol testing and personalized analysis. High-fidelity models, such as Ten Tusscher-Panfilov [2] or O’Hara-Rudy [3]

are accurate but computationally expensive. The Eikonal model [4] simulates periodic activation, making it unable to capture most arrhythmias. Intermediate models, such as FitzHugh-Nagumo [5] and Mitchell-Schaeffer (MS) [6] balance speed and complexity, yet still require significant computational time to solve PDEs using methods like Finite Element (FEM) [7] on realistic geometries.

Deep learning has emerged as a promising alternative for rapid PDE approximations [8][9], addressing computational bottlenecks. However, this approach faces challenges in patient-specific contexts due to variations in anatomy, pathology, and other factors. Differences in heart geometry and the distribution of damaged versus healthy tissue significantly affect propagation dynamics, requiring adaptable models capable of handling such variability. Graph Neural Networks (GNNs) [10] overcome these limitations by representing meshes as graphs [11]. However, these approaches typically require high-quality training data and often struggle with complex and irregular geometries like those found in medical applications.

We then propose **AGATA**, an Autoregressive Graph ATtention network for Action Potential, that uses advanced message-passing techniques [12] and attention mechanisms [13] to generate non-linear equations with multiple parameters. AGATA generates AP propagation over several seconds using just 25 ms of recorded data, providing a fully data-driven and generalizable approach. The model adapts to various tissue types and geometries, training on one type of synthetic mesh (e.g., triangular) while inferring on others (e.g., tetrahedral), thus addressing critical challenges in patient-specific cardiac modeling.

2 Methods

In this section, we convert the simulated AP propagation from a mesh to a graph representation, more flexible for machine learning, and allowing direct incorporation of geometric information in the dataset. We then introduce AGATA, a GNN-based architecture designed to mirror the mathematical dynamics of AP propagation as described by the MS model. We chose to use the MS model because it captures essential excitation and recovery dynamics. This makes it a solid basis for our future work, where we aim to extend AGATA to simulate arrhythmias and non-periodic activations.

2.1 Dataset Representation

A schema presenting the data generation framework is presented in Figure 1.

Mesh Generation: We used an ellipsoid mesh to represent a simplified left ventricle, generated with MMG [14]. We define regions in the ellipsoid varying in size and position, representing Scars and Gray Zones. The Gray Zone represents areas of slowed AP propagation, while there is no propagation in the scar region.

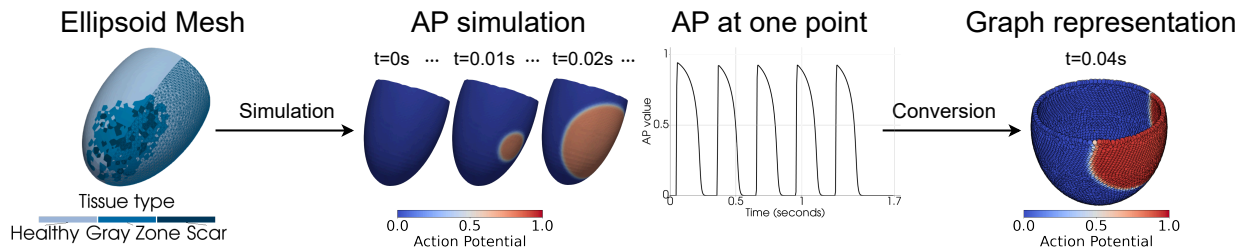


Figure 1: Framework for AP simulation, starting from an ellipsoid mesh, with Firedrake FEM computations and Pytorch Geometric (PyG) graph conversion. t is a step of the simulation.

Mathematical Model: To simulate the propagation of the AP, we used the MS model as presented by Djabella et al [15]. The voltage, which is dimensionless and scaled so that it ranges between zero and one is given by :

$$\begin{aligned}\partial_t u &= \overbrace{\text{div}(\mathbf{D}\nabla u)}^{\text{Diffusion}} + \overbrace{\frac{hu(u-\lambda)(u_{max}-u)}{\tau_{in}} - \frac{u}{\tau_{out}} + J_{stim}}^{\text{Reaction}}, \\ \partial_t h &= \begin{cases} \frac{1-h}{\tau_{open}} & \text{if } u < u_{gate}, \\ \frac{-h}{\tau_{close}} & \text{if } u > u_{gate}, \end{cases}\end{aligned}\tag{1}$$

where u is the time-dependent normalized dimensionless transmembrane potential variable, h is a gating variable that controls the repolarization, and J_{stim} is the stimulation current. Other parameters are described in [6].

To numerically solve the equations, we used an implicit-explicit scheme, as explained in [16]. We implemented it in Firedrake [17], a fast and well-documented PDE solver using FEM, chosen for its efficiency and near-mathematical syntax.

Graph Representation: The FEM solution was converted into a Pytorch-Geometric [18] graph. The node features correspond to the AP values over time, while the edge features encode the edge length and tissue properties.

2.2 Neural Network Architecture

AGATA is a GNN-based autoregressive model with node and edge attention mechanisms. The diffusion can be represented as a convolution, used in attention mechanisms, while the reaction is handled by the autoregressive architecture. The full architecture is depicted in Figure 2.

For a simulation of T steps, the mesh X is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes $i \in \mathcal{V}$ have features $u_i \in \mathbb{R}^T$ corresponding to the AP values, and each edge $ij \in \mathcal{E}$ encodes the edge length and tissue type as features $e_{ij} \in \mathbb{R}^3$.

Autoregressive process: AGATA predicts multiple future time steps simultaneously using a sliding window approach, inspired by Brandstetter et al. [19]. Temporal bundling enables the model to use a past historical time windows of size $\mathcal{T}_w$ for future predictions, enhancing stability and computational efficiency.

Graph Attention layer: Attention mechanism in graphs is a Message Passing process that assigns importance weights to the neighborhood of a node, as shown by Velickovic et al. [20]. We use GATv2Conv [21] layers, a dynamic attention mechanism. The attention weight between two nodes i and j is calculated as follows:

$$\alpha_{ij} = \frac{\exp(a^\top \text{LeakyReLU}(W_n(\mathbf{x}_i^s + \mathbf{x}_j^s) + W_e e_{i,j}))}{\sum_{k \in \mathcal{N}_i \cup \{i\}} \exp(a^\top \text{LeakyReLU}(W_n(\mathbf{x}_i^s + \mathbf{x}_k^s) + W_e e_{i,k}))}\tag{2}$$

where $\mathbf{x}_i^s$ is the new temporary updated feature of the node in the sliding window s , $\mathcal{N}(i)$ is the neighborhood of node i ; a , W_n and W_e are learnable weights for nodes and edges, respectively, and W global learnable weights. After several tests, three GATv2Conv layers provided a balance between computational efficiency and predictive accuracy.

Lastly, we concatenate all the windows outputs (see Fig. 2) and apply a linear layer and a sigmoid function to smooth and rescale the final output.

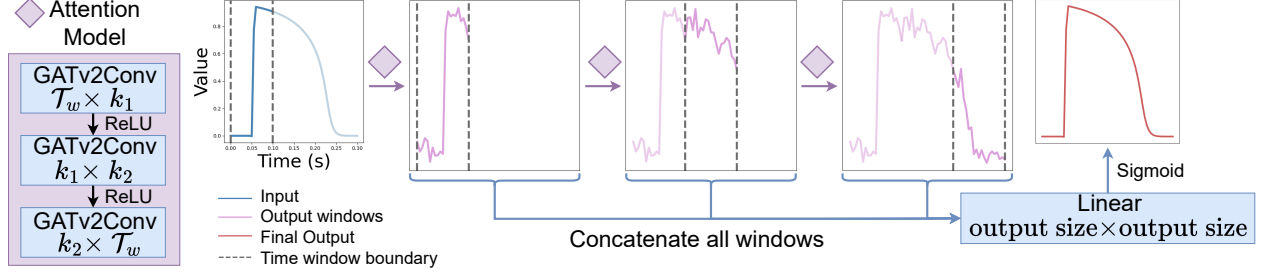


Figure 2: The architecture of AGATA, combining an autoregressive approach with an attention mechanism. k_1 , k_2 are the number of nodes in the layers.

3 Experiments

This section covers the data generation process and training results on the ellipsoid mesh. We then evaluate AGATA’s performance on more complex meshes, including real anatomical geometries. A key challenge is training AGATA on synthetic ellipsoid meshes while generalizing to volumetric meshes, including real anatomical cases with limited data, enabling inference without retraining.

3.1 Data Generation

Training requires a large quantity of data representative of the target data distribution. Since real and realistic data are scarce, we trained our model on simulation data using a simplified setting. More precisely, we used a single ellipsoid mesh with 5,360 points and 10,628 triangular elements. Random Scars and Gray Zones were generated as described previously, with one stimulus location per simulation. Each simulation applied five stimuli (5 ms each, spaced 300 ms apart) over 1.7 seconds. The simulation was performed with a 0.25 ms time step.

Parameter values for Healthy tissue were: $\tau_{in} = 0.0003$, $\tau_{out} = 0.006$, $\tau_{close} = 0.093$, $\tau_{open} = 0.12$, $u_{gate} = 0.13$, $u_{max} = 1$, $\lambda = 0.01$, $D = 1$. For Gray Zone: $\tau_{in} = 0.0003$, $\tau_{out} = 0.006$, $\tau_{close} = 0.235$, $\tau_{open} = 0.12$, $u_{gate} = 0.13$, $u_{max} = 0.9$, $\lambda = 0.2$, $D = 0.3$. All parameters, as explained in [6], are in seconds, except u_{max} and u_{gate} (dimensionless) and D (m^2/s). A total of 5,360 unique simulations were created as GT for AGATA training and evaluation. Datasets were split by stimulation location: 12.5% for validation and test (first quadrant: $x > 0$, $y > 0$), with 75% used for training.

3.2 Training

We chose the Root Mean Square Error (RMSE) loss to accentuate small differences, particularly in repolarization dynamics. Additionally, we report absolute errors and Mean Absolute Errors (MAE). Since AP values are dimensionless, all these error metrics are also dimensionless. We used a time window size of $T_w = 5$ steps, which corresponds to 25 ms. It is the smallest input size retaining enough temporal information for accurate propagation modeling. We performed hyperparameter tuning using Bayesian optimization (Optuna [22]), optimizing parameters such as the learning rate (lr), batch size, weight decay, dropout rate, and number of nodes per layer (k_1 , k_2). The best configuration resulted in a learning rate of 2.9×10^{-4} , a batch size of 2, a weight decay of 1.5×10^{-7} , a dropout rate of 0.335, $k_1 = 41$, and $k_2 = 36$. We used the Adam optimizer and obtained a training loss of 0.047 and a validation loss of 0.045. The different trials are summarized in Figure 3.

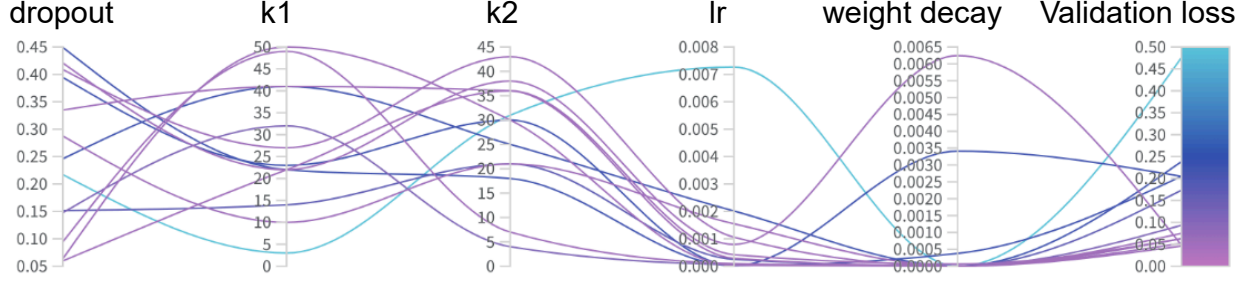


Figure 3: Different hyperparameters combinations using Bayesian optimization.

Qualitative results, (Figure 4), generated in Paraview [23], demonstrate the strong performance of AGATA. The propagation closely matches the GT, with well-preserved values and activation patterns, even in regions with complex wavefront interactions.

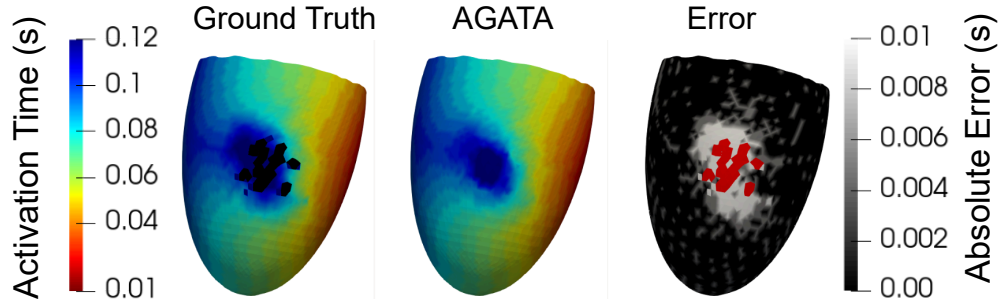


Figure 4: The Activation Map of the first stimulus, on the ellipsoid mesh, in seconds (s). Scars are represented in black in the Activation Map (left and middle panels) and in red in the Error Map (right panel). The Mean Absolute Error is 9×10^{-4} s with a standard deviation of 0.002 s.

As shown in Figure 5, in Healthy tissue, the depolarization occurs instantaneously, and the repolarization closely aligns with the GT.

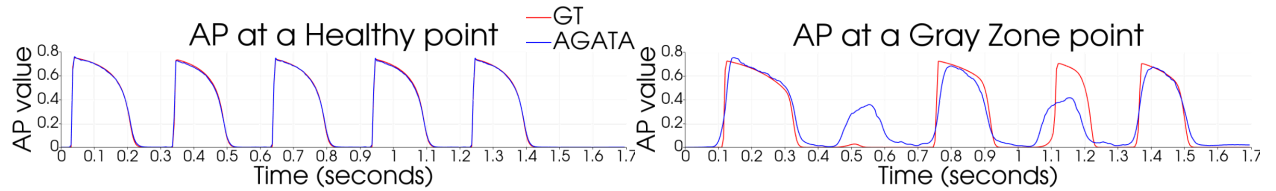


Figure 5: Temporal evolution of the AP at two representative points on the ellipsoid mesh.

AGATA achieves a MAE of 0.009 across the full test dataset, with a standard deviation (std) of 0.025 in Healthy tissue. For the Gray Zone, although some shape differences are observed, the numerical values remain accurate, such as the frequency of depolarization, with a MAE of 0.042 and a std of 0.071. Figure 6 shows a similar global propagation between GT and AGATA. The overall MAE is 0.016 and a std of 0.046, highlighting AGATA's consistency across different regions of the mesh.

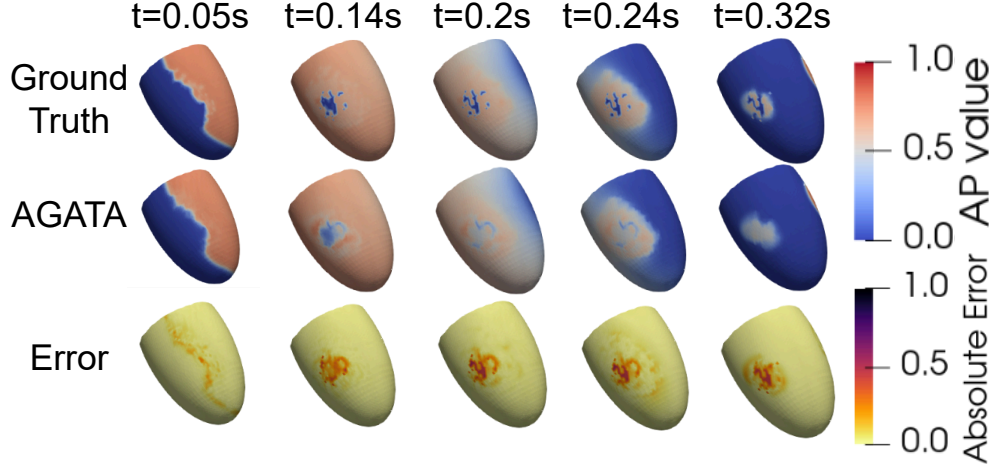


Figure 6: Comparison of spatial AP propagation on the ellipsoid mesh at specific time step.

These results validate the model’s ability to capture propagation dynamics on synthetic meshes. We will now evaluate its potential for application to more complex and clinically relevant geometries.

3.3 Generalization

AGATA was applied to two distinct real-world meshes, a 2D triangular mesh (Endocardium) and a 3D tetrahedron mesh (Ventricles), demonstrating its ability to generalize across different mesh types without retraining.

Endocardium: The left ventricle endocardium (LV-E) mesh is from a retrospective dataset provided by the University Hospital of Bordeaux, collected between 2010 and 2020 as described in [24]. This mesh is composed of 4999 points and 9917 triangular cells. We used the same parameters and solver as in previous simulations to model the electrical propagation.

As shown in Figure 7, AGATA predicts a slightly earlier activation compared to the GT, leading to a temporal shift in the effective refractory period.

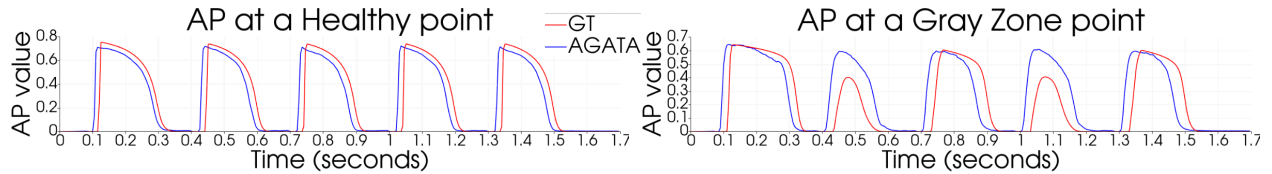


Figure 7: Temporal evolution of the AP at two representative points on the 2D LV-E mesh.

However, despite this temporal shift, the overall conduction pathway remains consistent with the GT, as illustrated in Figure 8.

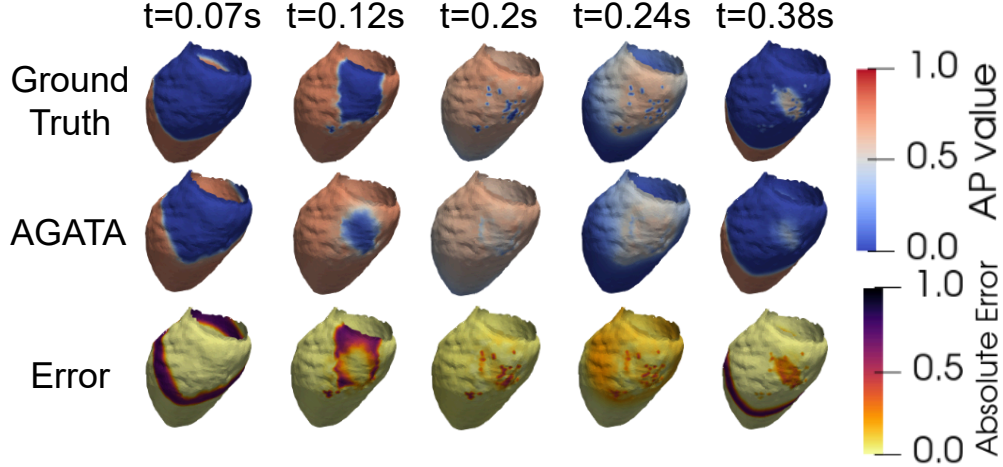


Figure 8: Comparison of spatial AP propagation on the 2D LV-E mesh at specific time step.

The spatial propagation pattern aligns well, indicating that AGATA effectively captures the global wavefront dynamics, as shown in Figure 9.

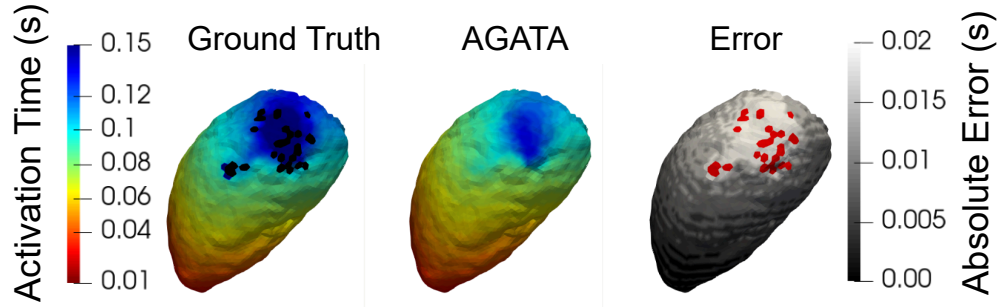


Figure 9: The Activation Map of the first stimulus, on the 2D LV-E mesh, in seconds (s). Scars are represented in black in the Activation Map (left and middle panels) and in red in the Error Map (right panel). The Mean Absolute Error is 8×10^{-3} s with a standard deviation of 0.005 s.

Ventricles: The ventricular mesh [25] consists of 31,330 tetrahedrons and 7,210 points. This case is particularly relevant, as AGATA was not trained on 3D geometries or volumetric boundaries such as the ventricular apex.

As shown in Figure 10, the wavefront shows only a slight local delay, while the overall signal shape is preserved.

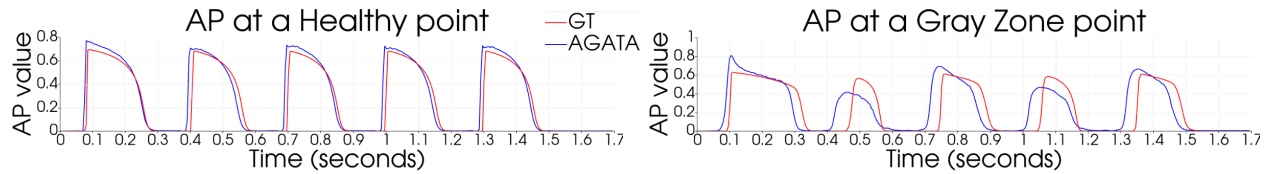


Figure 10: Temporal evolution of the AP at two representative points on the 3D ventricular mesh.

Figure 11 highlights minor deviations at the top of the ventricles.

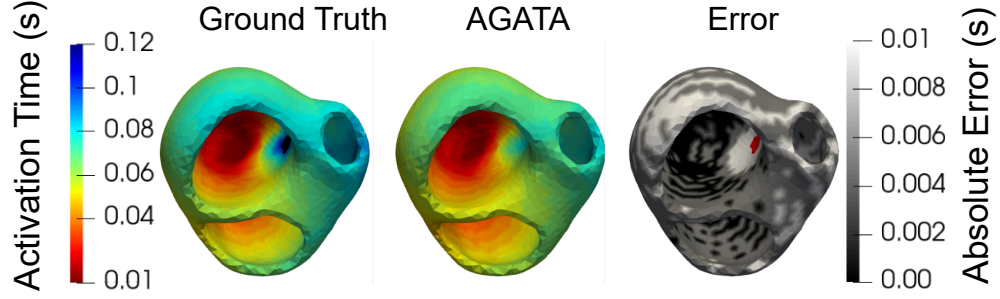


Figure 11: Activation map of the first stimulus on the 3D ventricular mesh (in seconds). Scars appear in black in the activation maps (left and middle panels) and in red in the error map (right panel). The mean absolute error is 4×10^{-3} s with a standard deviation of 0.003 s.

Results in Figure 12 demonstrate that AGATA generalizes well to 3D meshes despite geometric differences, preserving the global conduction pathway.

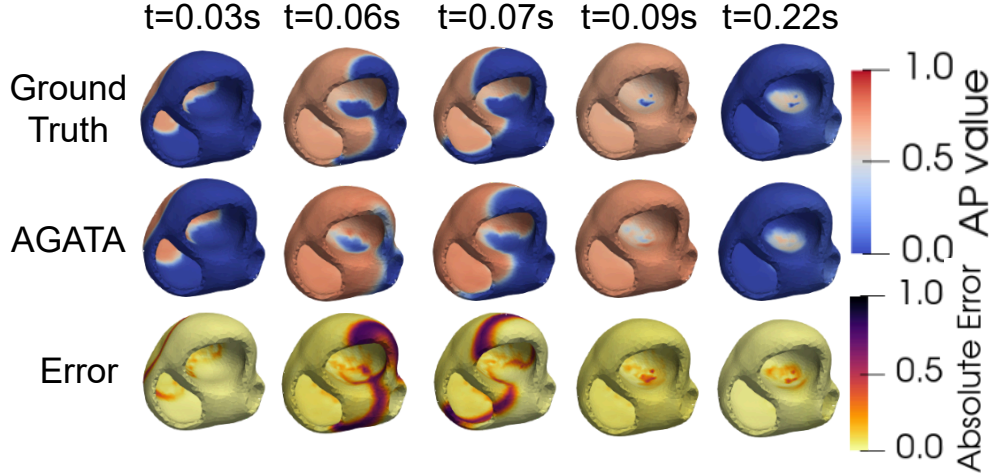


Figure 12: Comparison of spatial AP propagation on the 3D ventricular mesh at a specific time step.

Quantitative results: As shown in Table 1, the slightly larger MAE in LV-E and ventricular meshes are attributed to AGATA’s temporal advancement in propagation. The most important difference is in the Scar regions that are under-represented in the simulations. Despite these minor deviations, the overall patterns are similar, and the errors are negligible given that the AP range is between 0 and 1, highlighting AGATA’s robust performance across mesh types.

Table 1: Mean Absolute Error (with standard deviation) of overall AP propagation, in each different tissue type and globally.

Mesh	Mean Absolute Error (Standard Deviation)			
	Healthy	Gray Zone	Scar	Global
Ellipsoid	2.9e-05 (2.7e-5)	3.1e-5 (2.7e-5)	4.6e-5 (9.1e-5)	3.0e-5 (3.2e-5)
LV-E	0.007 (0.017)	0.016 (0.023)	0.024 (0.025)	0.008 (0.018)
Ventricles	0.005 (0.013)	0.008 (0.012)	0.015 (0.019)	0.006 (0.014)

The simulation times for AGATA and FEM are presented in Table 2. All conversions and FEM generation were done on a CPU 2.9GHz. AGATA’s inference was tested on the same CPU and on a Nvidia A40 GPU.

Table 2: Comparison of simulation times (seconds) between FEM and AGATA. In our case, Input is 25ms of simulation generated with FEM. Totals columns represent the time needed to generate 1.7s of propagation

	Input Generation	Mesh to Graph	Inference CPU/GPU	Graph to Mesh	Total AGATA CPU/GPU	Total FEM
Ellipsoid	18	12	3/0.5	16	49/46.5	302
LV-E	17	11	3/0.4	13	44/41.4	293
Ventricles	32	25	8/0.6	34	99/91.6	1159

AGATA significantly outperforms FEM in terms of computational efficiency. The largest gain is observed in the ventricular mesh, where it is over twelve times faster than FEM. This demonstrates AGATA’s ability to efficiently handle complex 3D geometries while maintaining accuracy.

Additionally, the current implementation is not fully optimized, suggesting that further improvements in simulation time are possible, notably when passing from one framework to another (e.g., mesh to graph).

4 Limitations

AGATA demonstrates robust performance on various mesh configurations, including 2D and 3D meshes, making it highly suitable for patient-specific applications. Certain limitations arise due to structural differences between train and test meshes. Figure 13 presents histograms comparing the edge lengths and connectivity (number of neighbors) for the ellipsoid, LV-E, and ventricular meshes.

These differences impact generalization. For example, while ellipsoids have more points than the LV-E, the latter represents a larger physical space. As a result, AGATA exhibits a faster propagation than expected. Similarly, the volumetric ventricular mesh exhibits larger average edge lengths and a higher variability in connectivity, that AGATA could not capture during training on a surface mesh. Another limitation is the under-representation of the Gray Zone and Scar regions in the training data, resulting in higher errors for these regions. Addressing these limitations may involve incorporating a greater diversity of training meshes, including a range of geometries and topologies, as well as more comprehensive combinations of Scar and Gray Zone tissue distributions.

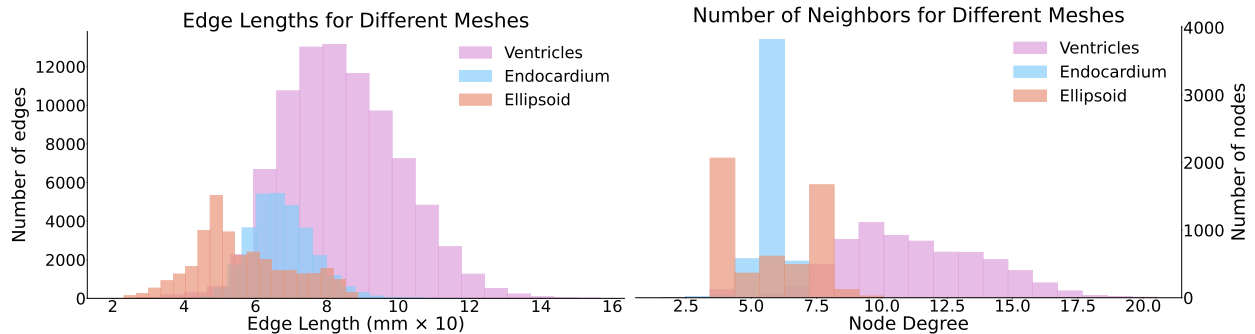


Figure 13: Histograms comparing the edge length (left) and the connectivity (right) of the ellipsoid (2D), endocardium (2D), and ventricles (3D) meshes. Node degree stands for the number of neighbors of a node.

This approach would enhance AGATA’s ability to generalize to complex and previously unseen cardiac geometries.

5 Conclusion

We introduced AGATA, a novel autoregressive graph attention neural network framework for simulating cardiac action potential propagation. Trained on simple synthetic surface meshes, AGATA demonstrated its ability to generalize to real and complex geometries, such as the left ventricle endocardium and a biventricular volumetric mesh. This adaptability highlights its potential for patient-specific applications without requiring retraining or fine-tuning. AGATA effectively captures the propagation patterns of cardiac electrophysiology, accurately modeling behaviors in healthy tissue and activation patterns in damaged areas. Furthermore, it achieves a significant reduction in computational time compared to traditional FEM, enabling near-real-time simulations and making it suitable for time-sensitive clinical and research applications. The current structure of AGATA makes it a promising candidate for simulating anisotropic propagation and arrhythmias. Its localized attention mechanism suggests it could naturally adapt to different propagation patterns across multiple dimensions, potentially handling variations in conduction velocities in 3D meshes. However, further validation is needed to confirm its effectiveness in these scenarios.

6 Disclosure of Interests.

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The authors have no competing interests to declare that are relevant to the content of this article.

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